

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2025 4WM2H Worked Solutions

May2025 | 4WM2H | Unit 2

35 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 1** Rio has six cards.
He writes a number on each card so that
- the range of the numbers is 10
 - the median of the numbers is 7.5
 - the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14
--	--	---	--	----	----

Three of the numbers are hidden.
Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 2 The diagram shows a solid prism with a cross section in the shape of a trapezium.

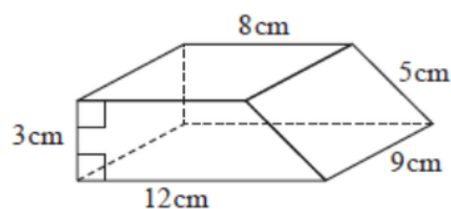


Diagram NOT
accurately drawn

Work out the total surface area of the prism.

..... cm²

(Total for Question 2 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 3

TOPIC: **Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 3**TOPIC: **Percentages**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 3** Bernice bought a car for \$8000
She sold the car for \$5600

Work out Bernice's percentage loss when she sold the car.

.....%

(Total for Question 3 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 3

Marks: 3

TOPIC: **Percentages**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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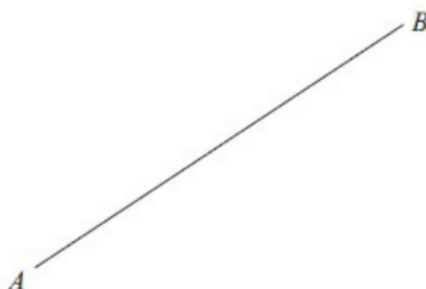
No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 4** Using ruler and compasses only, construct the perpendicular bisector of the line AB .
Show all your construction lines.

**(Total for Question 4 is 2 marks)**

EA

PAST PAPER SOLUTION

Q#: 4

Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 5****Marks: 3****TOPIC: Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 5** Here are the first five terms of a number sequence.

11 15 19 23 27

- (a) Find an expression in terms of n for the n th term of this sequence.

.....
(2)

The n th term of another number sequence is $5n + 4$

Salim says no term in this sequence is a prime number.

- (b) Find a term in this sequence that shows Salim is wrong.

.....
(1)

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5

Marks: 3

TOPIC: **Sequences**

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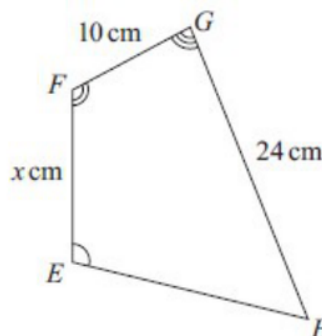
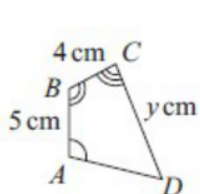
No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 6** **Marks: 4****TOPIC:** Congruence, Similarity & Geometrical Proof

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 6 The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

Diagram **NOT**
accurately drawn

$$\begin{array}{lll} AB = 5 \text{ cm} & BC = 4 \text{ cm} & CD = y \text{ cm} \\ EF = x \text{ cm} & FG = 10 \text{ cm} & GH = 24 \text{ cm} \end{array}$$

- (a) Work out the value of x

$$x = \dots\dots\dots (2)$$

- (b) Work out the value of y

$$y = \dots\dots\dots (2)$$

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6

Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Ratio Problem Solving**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 7 Pau, Sam and Tia share £240 in the ratios 3 : 4 : 5

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.

Give your answer in its simplest form.

..... : :

(Total for Question 7 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 7

Marks: 4

TOPIC: **Ratio Problem Solving**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 4****TOPIC: Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 8** The table gives information about the lengths of time, in minutes, that some people were in a coffee shop.

Length of time (m minutes)	Frequency
$0 < m \leq 10$	x
$10 < m \leq 20$	3
$20 < m \leq 30$	2
$30 < m \leq 40$	4
$40 < m \leq 50$	2

The information in the table is used to calculate an estimate for the mean length of time these people were in the coffee shop.

The estimate for the mean length of time is 20

Work out the value of x

$x = \dots\dots\dots$

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8

Marks: 4

TOPIC: **Statistics Toolkit**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Compound Interest & Depreciation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 9** Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.
Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

..... Swiss francs

(Total for Question 9 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 9

Marks: 3

TOPIC: **Compound Interest & Depreciation**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Simultaneous Equations**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

10 Solve the simultaneous equations

$$\begin{aligned} 3x + 5y &= 8 \\ 4x + y &= -3.5 \end{aligned}$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$ **(Total for Question 10 is 3 marks)**

EA

PAST PAPER SOLUTION

Q#: 10

Marks: 3

TOPIC: **Simultaneous Equations**

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EA

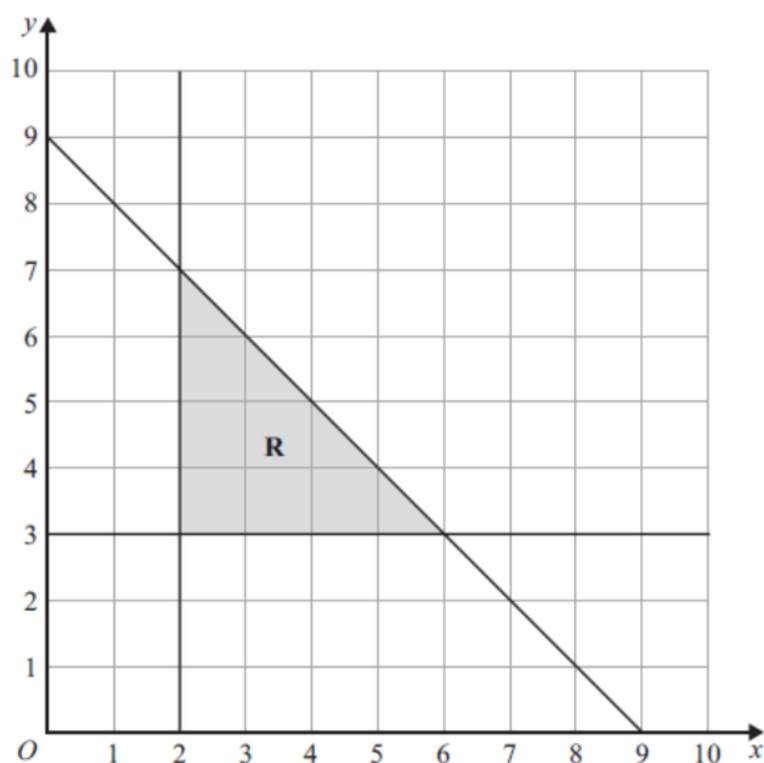
PAST PAPER QUESTION**Q#: 11** **Marks: 5****TOPIC: Graphing Inequalities**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 11** (a) Solve the inequality $7 - 3t < 2t + 15$

.....
(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down three inequalities that define the region **R**

.....
.....
.....
(3)**(Total for Question 11 is 5 marks)**

PAST PAPER SOLUTION

Q#: 11

Marks: 5

TOPIC: **Graphing Inequalities**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 12** (a) Write 8.6×10^{-5} as an ordinary number.

.....
(1)

- (b) Work out $(3 \times 10^{143}) \div (6 \times 10^{137})$

Give your answer in standard form.

.....
(2)**(Total for Question 12 is 3 marks)**

EA

PAST PAPER SOLUTION

Q#: 12

Marks: 3

TOPIC: **Powers, Roots & Standard Form**

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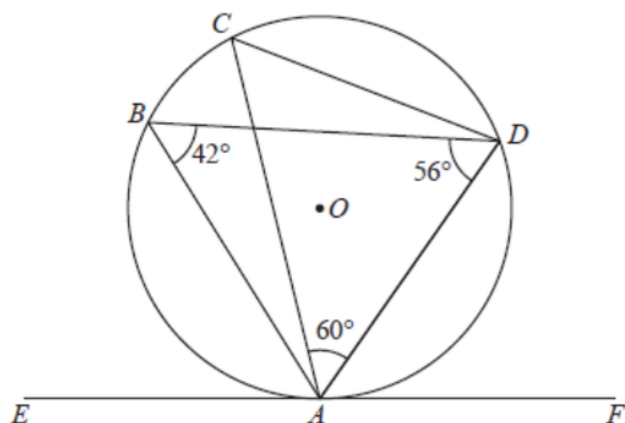
No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 13** **Marks: 4****TOPIC: Circle Theorems**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 13 The diagram shows a circle, centre O

Diagram NOT
accurately drawn

A, B, C and D are points on the circle.
 EAF is the tangent to the circle at A

angle $ABD = 42^\circ$ angle $BDA = 56^\circ$ angle $CAD = 60^\circ$

- (a) (i) Find the size of angle EAB

.....^o
 (1)

- (ii) Give a reason for your answer to (a) (i)

.....

 (1)

- (b) Work out the size of angle CDB

.....^o
 (2)

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION

Q#: 13

Marks: 4

TOPIC: **Circle Theorems**

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PAST PAPER QUESTION**Q#: 14** **Marks: 6****TOPIC: Cumulative Frequency Diagrams**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 14 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

Time (t minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	13
$20 < t \leq 30$	12
$30 < t \leq 40$	17
$40 < t \leq 50$	7
$50 < t \leq 60$	3

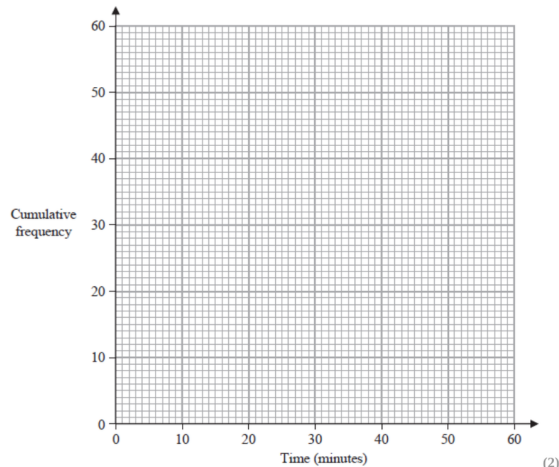
- (a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$10 < t \leq 20$	
$20 < t \leq 30$	
$30 < t \leq 40$	
$40 < t \leq 50$	
$50 < t \leq 60$	

(1)

- (b) On the grid on the next page, draw a cumulative frequency graph for your table.

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(2)

- (c) Use your graph to find an estimate for the median time.

..... minutes
(1)

- (d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

.....
(2)

(Total for Question 14 is 6 marks)

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Q#: 14

Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

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PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Compound Interest & Depreciation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

15 Fred buys a painting.

The value of the painting increases by 12% during the first year.

The value of the painting then decreases by 15% during the second year.

Work out the overall percentage decrease in the value of the painting during this two-year period.

.....%

(Total for Question 15 is 3 marks)

EA

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Q#: 15

Marks: 3

TOPIC: **Compound Interest & Depreciation**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 16****Marks: 2**TOPIC: **Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

16 Here are the numbers of goals scored by a netball team in its 11 games this season.

7 8 10 11 13 14 16 17 19 21 24

Work out the interquartile range for the number of goals.

.....
(Total for Question 16 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 16

Marks: 2

TOPIC: **Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

17 $A = 3^2 \times 5^4$ $B = 2^5 \times 3^2$ $C = 2^3 \times 5^6$

- (a) Work out the value of $(ABC)^3$

Give your answer as a product of powers of its prime factors.

.....
(2)

- (b) Work out $(5 \times 10^{150}) \times (3 \times 10^{140})$

Give your answer as a product of powers of its prime factors.

.....
(2)

(Total for Question 17 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 17

Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 3****TOPIC: Direct & Inverse Proportion**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

18 W is inversely proportional to y^2 $W = 50$ when $y = 4$ Find a formula for W in terms of y

.....

(Total for Question 18 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 18

Marks: 3

TOPIC: **Direct & Inverse Proportion**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 4****TOPIC: Rearranging Formulas**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

19 Make t the subject of the formula $u = \sqrt{7 - 8t^2}$

.....
(Total for Question 19 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 19

Marks: 4

TOPIC: **Rearranging Formulas**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 3**TOPIC: **Algebraic Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

20 | a, b and c are three consecutive even numbers where $a < b < c$ Prove algebraically that $b^2 = ac + 4$ **(Total for Question 20 is 3 marks)**

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Algebraic Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 3**TOPIC: **Algebraic Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

20 | a, b and c are three consecutive even numbers where $a < b < c$ Prove algebraically that $b^2 = ac + 4$ **(Total for Question 20 is 3 marks)**

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Algebraic Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 7****TOPIC: Functions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

21 The function g is defined as

$$g(x) = \frac{7x + 20}{2x} \quad x \neq 0$$

- (a) Solve $g(x) = 6$
Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

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The function h is defined as

$$h(x) = 5x^2 + 30x - 7 \quad x \geq -3$$

- (b) Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots (4)$$

(Total for Question 21 is 7 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 7

TOPIC: **Functions**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 21****Marks: 7****TOPIC: Functions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

21 The function g is defined as

$$g(x) = \frac{7x + 20}{2x} \quad x \neq 0$$

- (a) Solve $g(x) = 6$
Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

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$$h(x) = 5x^2 + 30x - 7 \quad x \geq -3$$

- (b) Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots (4)$$

(Total for Question 21 is 7 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 7

TOPIC: **Functions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Differentiation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 22** Work out the range of values of x for which the graph of $y = x^3 + \frac{1}{2}x^2 - 10x + 5$ has a positive gradient.
Show clear algebraic working.

.....

(Total for Question 22 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Differentiation**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Differentiation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 22** Work out the range of values of x for which the graph of $y = x^3 + \frac{1}{2}x^2 - 10x + 5$ has a positive gradient.
Show clear algebraic working.

.....

(Total for Question 22 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Differentiation**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 23** **Marks: 3****TOPIC: Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 23 The diagram shows a cuboid.

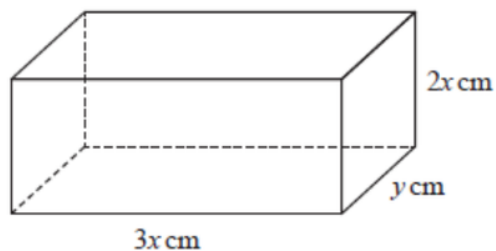


Diagram NOT
accurately drawn

The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A \text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 23 is 3 marks)

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Q#: 23

Marks: 3

TOPIC: **Volume & Surface Area**

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PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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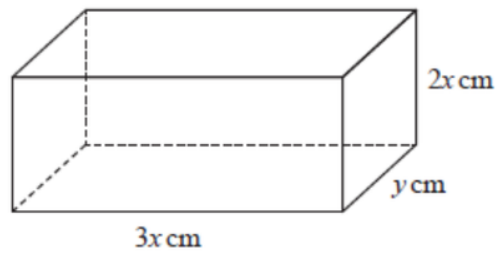


Diagram NOT
accurately drawn

The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A \text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 3

TOPIC: **Volume & Surface Area**

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PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 24** The 21st term of an arithmetic series is 109
The sum of the first 52 terms of the series is 4381
Work out the 5th term of the series.
Show clear algebraic working.

.....
(Total for Question 24 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 4

TOPIC: **Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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EA

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 24** The 21st term of an arithmetic series is 109
The sum of the first 52 terms of the series is 4381
Work out the 5th term of the series.
Show clear algebraic working.

.....
(Total for Question 24 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 4

TOPIC: **Sequences**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 25****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

25 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 25 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 25

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 25****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 25 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 25

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

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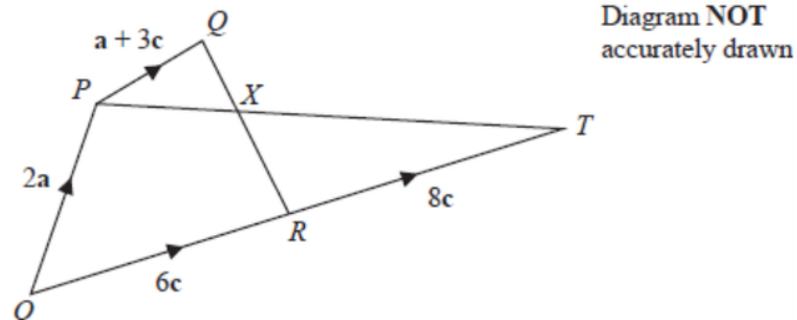
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EA

PAST PAPER QUESTION**Q#: 26****Marks: 5****TOPIC: Vectors**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

26

The diagram shows a quadrilateral $OPQR$ in which

$$\vec{OP} = 2\mathbf{a} \quad \vec{OR} = 6\mathbf{c} \quad \vec{PQ} = \mathbf{a} + 3\mathbf{c}$$

- (a) Find $\vec{QR}$ in terms of $\mathbf{a}$ and $\mathbf{c}$

$$\vec{QR} = \dots\dots\dots (1)$$

The point T is such that $\vec{RT} = 8\mathbf{c}$
 The point X lies on QR such that PXT is a straight line.

- (b) Using a vector method, find the ratio $QX : XR$
 Give your answer in the form $m : n$ where m and n are integers.
 Show your working clearly.

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$$QX : XR = \dots\dots\dots (4)$$

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26

Marks: 5

TOPIC: **Vectors**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

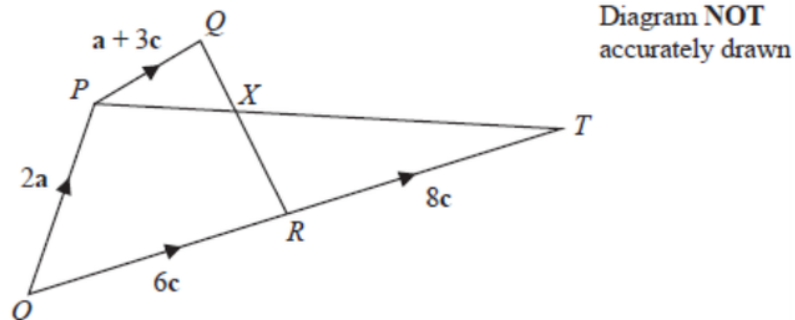
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 26****Marks: 5****TOPIC: Vectors**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

26

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Q#: 26

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TOPIC: **Vectors**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 27****Marks: 4****TOPIC: Transformations of Graphs**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

- 27 A curve has equation
- $y = f(x)$

There is one maximum point on this curve.

The coordinates of this maximum point are $(6, -5)$

- (a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 4)$

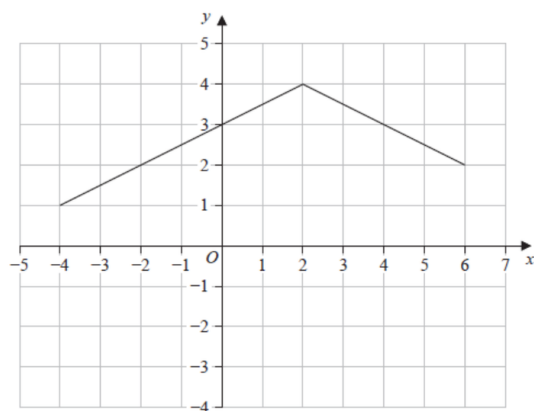
(.....,)
(1)

(ii) $y = f(3x)$

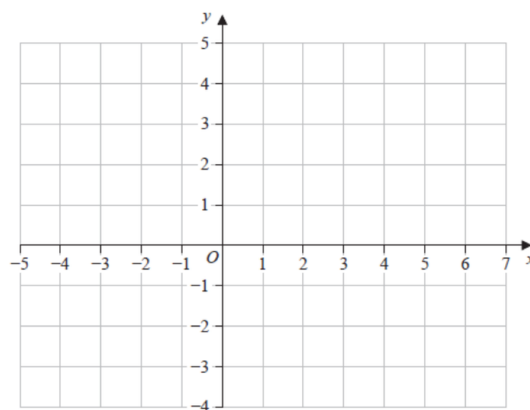
(.....,)
(1)

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Here is the graph of $y = g(x)$ 

- (b) On the grid below, draw the graph of
- $y = g(2x) - 3$



(2)

(Total for Question 27 is 4 marks)

PAST PAPER SOLUTION

Q#: 27

Marks: 4

TOPIC: **Transformations of Graphs**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 27****Marks: 4****TOPIC: Transformations of Graphs**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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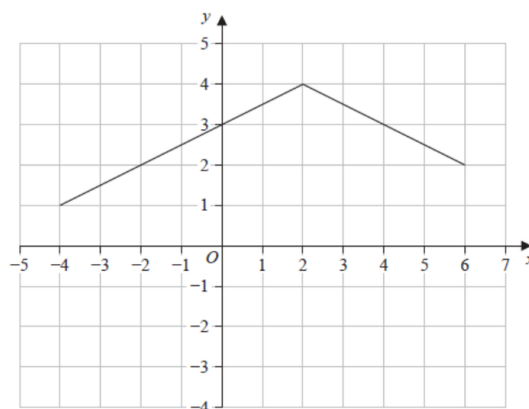
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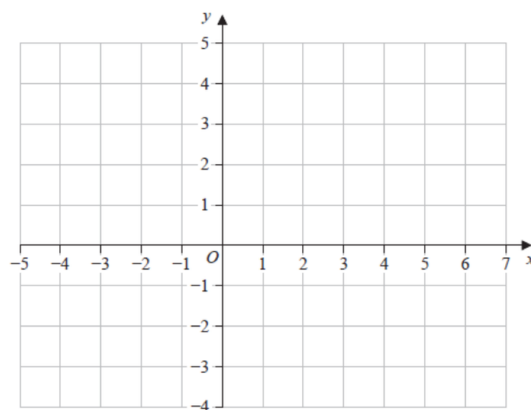
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TOPIC: **Transformations of Graphs**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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Dr. Eslam Ahmed

No worked solution is saved yet.

EA